



SAPIENZA
UNIVERSITÀ DI ROMA

Collection of statements

MIP Modelling — the problems of the course,
with no solution

The same problems as the notes, in the same order, with the texts only: to practise before reading the solution. For each of them you are asked for the model, the model of the instance, a constructive heuristic with its bound, and the dual of the LP relaxation with a feasible dual solution.

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Contents

I	Modelling	2
1	Logic and binary variables	3
2	Relaxations, duality and bounds	5
II	The problems	6
3	Fifteen numerical models	7
4	Assignment and scheduling	11
5	Location and coverage	13
6	Production planning	15
7	Mixed models	16



Part I

Modelling



Logic and binary variables

direct implications

Ten projects. Model as linear constraints:

1. if project 2 is chosen then project 3 is chosen;
2. if project 2 is chosen then project 4 is not chosen;
3. if projects 1 and 6 are chosen then project 7 is chosen;
4. if project 1 or project 6 is chosen then project 8 is chosen;
5. if projects 2 and 3 are chosen then project 9 is not chosen;
6. if project 2 or project 3 is chosen then project 10 is not chosen.

negated antecedents and consequents

Ten projects. Model:

1. if project 3 is not chosen then project 2 is chosen;
2. if project 4 is not chosen then project 2 is not chosen;
3. if project 7 is chosen then projects 1 and 6 are chosen;
4. if project 8 is chosen then project 1 or project 6 is chosen;
5. if project 9 is not chosen then projects 2 and 3 are chosen;
6. if project 10 is not chosen then project 2 or project 3 is chosen.

compound antecedents and consequents

Eight projects. Model:

1. $x_7 \text{ OR } x_3 \Rightarrow x_1 \text{ AND } x_2$;
2. $x_1 \text{ AND } x_6 \text{ AND } x_7 \Rightarrow x_8$;
3. $x_5 \text{ AND } x_2 \text{ AND NOT } x_4 \Rightarrow \text{NOT } x_3$;

4. $(x_1 \text{ OR } x_4) \text{ AND } x_6 \Rightarrow x_2 \text{ AND } (x_5 \text{ OR } x_7)$;
5. $(x_2 \text{ OR } x_5) \text{ AND } \text{NOT } x_8 \Rightarrow x_3 \text{ OR } \text{NOT } x_6$;
6. $(x_1 \text{ OR } x_4) \text{ AND } (x_2 \text{ OR } x_5) \text{ AND } \text{NOT } x_8 \Rightarrow x_3 \text{ AND } (\text{NOT } x_6 \text{ OR } x_7)$.

"at least two of"

Nine projects. Model:

1. if project 4 is chosen then at least two of 1, 2, 3;
2. if at least two of 6, 7, 8 then project 5;
3. if project 4 is not chosen then at least two of 1, 2, 3, 9;
4. if project 8 is chosen then (1 and 6) or (1 and 7) or (2 and 6);
5. if at least two of 1, 3, 5 then project 9 is not chosen;
6. if (1 and 2) or (3 and 4) then project 5.

splits

Ten projects. Model:

1. $x_1 \text{ OR } x_2 \Rightarrow x_3$;
2. $x_4 \Rightarrow x_5 \text{ AND } x_6$;
3. $x_1 \text{ OR } x_2 \Rightarrow x_3 \text{ AND } x_4$;
4. $x_1 \text{ AND } x_2 \Rightarrow x_3$;
5. $x_5 \text{ OR } x_6 \Rightarrow \text{NOT } x_7$;
6. $\text{NOT } x_8 \text{ OR } \text{NOT } x_9 \Rightarrow x_{10}$.

Relaxations, duality and bounds

Minimum-cost zone covering

A town has $n \in \mathbb{Z}_{\geq 1}$ districts; each has an emergency team, and activating the team of district $j \in \{1, 2, \dots, n\}$ costs $c_j \in \mathbb{Q}_{>0}$ thousand euros. There are $m \in \mathbb{Z}_{\geq 1}$ sensitive zones: zone $i \in \{1, 2, \dots, m\}$ is covered if at least one of the teams in the set S_i of the districts bordering it is active. All zones must be covered at minimum cost.

Knapsack

There are $n \in \mathbb{Z}_{\geq 1}$ items: item $j \in \{1, 2, \dots, n\}$ has value $p_j \in \mathbb{Q}_{>0}$ and weight $w_j \in \mathbb{Q}_{>0}$, and the knapsack has capacity $b \in \mathbb{Q}_{>0}$. Choose the subset of maximum value that fits.

Part II

The problems



Fifteen numerical models

EX 1 — which groups of tourists to accept

A tour operator rents an eight-seat van to guide tourists around a scenic location. On a given day four groups show up, of 2, 3, 4 and 5 people. Since the van cannot take all of them at once, the operator asks each group (whose members do not want to be separated) to make an offer: the four groups offer 30, 50, 80 and 70 euros. For operational reasons at most two groups can be accepted; moreover, if group 2 is accepted then group 4 must be accepted as well. The goal is to maximise the total earnings.

EX 2 — assignment with a capacity in number of lines

A city council is evaluating the offers of three bus companies for four bus lines. Each company has stated the cost at which it would operate each line:

Company	Line 1	Line 2	Line 3	Line 4
1	10	4	9	7
2	1	2	3	10
3	8	9	10	1

Every line must be assigned to exactly one company and every company may operate at most two lines. The total cost is to be minimised.

EX 3 — assignment with more resources than tasks

A coach must form a relay team by choosing three swimmers out of four. The race has three 50-metre legs: backstroke, breaststroke and butterfly. The recorded times (seconds) are:

Stroke	Carlo	Paolo	Fabio	Marco
Backstroke	37	32	33	37
Breaststroke	43	33	42	34
Butterfly	33	28	38	30

The total relay time is to be minimised.

EX 4 — production and workforce over three months

Over the next three months a shoe factory must meet monthly demands of 3000, 5000 and 7000 pairs of shoes. At the start of the first month it has 500 pairs in stock and employs 100 workers. Every worker is paid 1500 euros a month and works 160 hours; producing one pair requires 4 labour hours and 15 euros of raw materials. At the start of every month workers may be hired, at a cost of 100 euros each for training. Storing one unsold pair at the end of the month costs 3 euros. The factory wants the production and workforce plan of minimum total cost.

EX 5 — five types of vehicle

A company is considering the production of five vehicle types: compact car, midsize car, large car, midsize minivan and large minivan. It has 1000 tons of steel and 2000 labour hours available. The table reports, for every type, the steel per unit, the hours per unit, the unit profit and the minimum quantity to be produced if the type is selected:

Type	steel	hours	profit	minimum
compact car	2	30	200	10
midsize car	3	25	250	10
large car	5	40	300	10
midsize minivan	6	45	550	5
large minivan	8	55	700	5

The company wants the plan of maximum profit.

EX 6 — the minimum number of hubs

An airline organises its network with a *hub-and-spoke* system: a hub serves directly every city within 1000 miles. The company operates on eight cities; for each of them, the cities within 1000 miles are:

City	Cities within 1000 miles
Atlanta	Atlanta, Chicago, Houston, New York
Chicago	Atlanta, Chicago, New York
Denver	Denver, Los Angeles
Houston	Atlanta, Houston
Los Angeles	Denver, Los Angeles, San Francisco
New York	Atlanta, Chicago, New York
San Francisco	Los Angeles, San Francisco, Seattle
Seattle	San Francisco, Seattle

The minimum number of hubs is wanted, such that every city is within 1000 miles of at least one hub.

EX 7 — which orders to accept

A company builds custom-designed aircraft; before producing for a client a fixed set-up cost has to be paid. Three clients have ordered, respectively, 3, 2 and 5 aircraft. The production capacity is 1000 hours. The company decides which orders to accept and how many units to produce for each (possibly fewer than requested). Set-up costs, unit profits and hours per

aircraft:

	client 1	client 2	client 3
set-up (millions)	3	2	0
unit profit (millions)	2	3	1
hours per aircraft	200	400	200

The goal is to maximise the total profit net of the set-up costs.

EX 8 — selection with exact cardinality and non-adjacency

A student must attend *exactly two* sessions of a summer school. There are three seminars, each offered in four time slots (9–10, 10–11, 11–12, 12–13); every seminar is attended at most once and two sessions cannot occupy the same slot. The preference score of each session is:

	9–10	10–11	11–12	12–13
Seminar 1	8	6	5	3
Seminar 2	7	9	4	6
Seminar 3	5	7	8	9

The student never wants *two consecutive hours* of class. The total preference is to be maximised.

EX 9 — the largest number of queens

On an 8×8 chessboard one wants to place the largest number of queens so that no queen can capture another in a single move. In chess a queen moves any number of squares along the same row, the same column or either diagonal.

EX 10 — selection with a tool set

A CNC machine can perform five operations; each one requires a subset of six tools and, if performed, yields a profit:

Operation	Required tools	Profit (€)
1	1, 3, 4, 5	2000
2	1, 2, 6	1500
3	1, 2, 3, 5	1800
4	2, 4, 5	1700
5	5, 6	800

The magazine of the machine holds at most $k = 4$ tools at a time. The configuration of maximum profit is wanted.

EX 11 — workloads as balanced as possible

Four indivisible jobs of duration 2, 3, 6 and 7 must be assigned to two workers so that their workloads are as balanced as possible.

EX 12 — three types of shoe with a threshold

A shoe company produces hiking boots, loafers and walking shoes. The table reports the resources needed for one pair, the minimum quantity to be produced if the type is produced, and the selling price:

Type	leather (g)	hours	nails	minimum	price
hiking boots	850	3	20	100	150
loafers	600	2	15	200	120
walking shoes	700	2.5	20	150	130

There are 120 000 grams of leather, 7000 machine hours and 40 000 nails available. The goal is to maximise the revenue.

EX 13 — two mutual funds

After a lottery win, an investor must decide how to allocate 100 million lire. A friend suggests two types of mutual fund. Shares can only be bought in indivisible lots: one lot of the first type costs 12 million, one of the second 20 million. The estimated annual return is $1/6$ of the capital invested for the first type and 15% for the second. Moreover the number of lots of the second type cannot exceed 50% of the total number of lots bought. The goal is to maximise the expected annual return.

EX 14 — weekly shifts

In an emergency department the number of full-time staff members required is given for every day of the week:

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
staff	10	8	12	9	9	7	8

One full-time member can be replaced by two part-time members. The shift pattern is fixed: four consecutive full-time days, then one part-time day, then two rest days. The weekly cost of a worker depends on the days worked: 100 euros for every weekday (Monday to Friday), 110 for Saturday and 130 for Sunday; these costs are halved on part-time days. The goal is the weekly plan of minimum cost.

EX 15 — weekly timetable

A music school needs to build its weekly timetable. There are four afternoons (Monday, Tuesday, Wednesday and Thursday), each of three hours. 6 hours of guitar, 3 of violin, 2 of piano and 1 of harp must be placed. Every day must include lessons of at least two different instruments. Piano and harp lessons cannot be in consecutive hours of the same day, because of a conflict between the two teachers. Every teacher has preferences to be avoided, if possible: the guitar teacher prefers not to teach at hour 1 on Monday and Tuesday, the violin teacher at hour 2 on Wednesday and Thursday, the piano teacher at hour 3 on any day, the harp teacher on Tuesday. The goal is to minimise the number of hours that violate a preference.

Assignment and scheduling

Problem 7.1 — minimum-cost assignment

A company has to process $n \in \mathbb{Z}_{\geq 1}$ jobs on $k \in \mathbb{Z}_{\geq 1}$ machines. For each job $j \in \{1, 2, \dots, n\}$ and each machine $m \in \{1, 2, \dots, k\}$, the value $t_{jm} \in \mathbb{Q}_{>0}$ is the processing time in minutes and the value $c_{jm} \in \mathbb{Q}_{>0}$ is the cost in euros of processing job j on machine m . For each machine $m \in \{1, 2, \dots, k\}$, the value $a_m \in \mathbb{Q}_{>0}$ is the available processing time in minutes. Each machine handles one job at a time (no parallel processing). The company seeks a minimum-cost assignment of all jobs to the machines.

Problem 7.2 — machines with fixed cost

A company needs to process $n \in \mathbb{Z}_{\geq 1}$ jobs and has $k \in \mathbb{Z}_{\geq 1}$ machines. For each job $j \in \{1, 2, \dots, n\}$ and each machine $m \in \{1, 2, \dots, k\}$, the value $t_{jm} \in \mathbb{Q}_{>0}$ is the processing time in minutes of job j on machine m . For each machine $m \in \{1, 2, \dots, k\}$, the value $a_m \in \mathbb{Q}_{>0}$ is the availability in minutes and the value $c_m \in \mathbb{Q}_{>0}$ is the cost in euros if the machine is used. Each machine handles one job at a time. The company wants to assign all jobs to the machines minimising the cost of the machines used.

Problem 7.3 — job selection

A company can execute $n \in \mathbb{Z}_{\geq 1}$ jobs and has $k \in \mathbb{Z}_{\geq 1}$ machines. For each job $j \in \{1, 2, \dots, n\}$, the value $t_j \in \mathbb{Q}_{>0}$ is the processing time in minutes (the same on every machine) and the value $r_j \in \mathbb{Q}_{>0}$ is the revenue in euros if the job is executed. For each machine $m \in \{1, 2, \dots, k\}$, the value $a_m \in \mathbb{Q}_{>0}$ is the availability in minutes and the value $c_m \in \mathbb{Q}_{>0}$ is the cost in euros if the machine is used. Each machine handles one job at a time. The company wants to choose which jobs to execute, and on which machines, to maximise the profit: revenues of the executed jobs minus costs of the machines used.

Problem 7.4 — parallel jobs

A company needs to execute $n \in \mathbb{Z}_{\geq 1}$ jobs using $k \in \mathbb{Z}_{\geq 1}$ machines. For each job $j \in \{1, 2, \dots, n\}$ and each machine $m \in \{1, 2, \dots, k\}$, the value $t_{jm} \in \mathbb{Q}_{>0}$ is the processing time in minutes of job j on machine m . Each machine $m \in \{1, 2, \dots, k\}$ can execute at most $p_m \in \mathbb{Z}_{\geq 1}$ jobs, and executes them in parallel: all the jobs assigned to the same machine start together. The company wants to assign the jobs minimising the sum of the processing

times of the machines, where the processing time of a machine is the time of the longest job among those assigned to it.

Problem 7.5 — classes with setup

A company has $n \in \mathbb{Z}_{\geq 1}$ jobs that can be executed on a machine with an available processing time of $a \in \mathbb{Q}_{>0}$ minutes. For each job $j \in \{1, 2, \dots, n\}$, the value $t_j \in \mathbb{Q}_{>0}$ is the processing time in minutes and the value $r_j \in \mathbb{Q}_{>0}$ is the revenue in euros if the job is executed. The jobs are partitioned into $q \in \mathbb{Z}_{\geq 2}$ classes: for each class $c \in \{1, 2, \dots, q\}$, the set $\mathcal{J}_c \subseteq \{1, 2, \dots, n\}$ contains the jobs of the class. If the machine executes jobs of a class c , a setup cost of $f_c \in \mathbb{Q}_{\geq 0}$ euros is incurred and a setup time of $s_c \in \mathbb{Q}_{\geq 0}$ minutes is needed to prepare the machine. The machine cannot execute jobs in parallel. The company wants to choose the jobs to execute maximising the profit.

Problem 7.6 — completion bonus

A company has $n \in \mathbb{Z}_{\geq 1}$ jobs that can be executed on a machine with an available processing time of $a \in \mathbb{Q}_{>0}$ minutes. For each job $j \in \{1, 2, \dots, n\}$, $t_j \in \mathbb{Q}_{>0}$ is the processing time in minutes and $r_j \in \mathbb{Q}_{>0}$ the revenue in euros if executed. The jobs are partitioned into $q \in \mathbb{Z}_{\geq 2}$ classes $\mathcal{J}_1, \dots, \mathcal{J}_q$. The availability of the machine is reduced by $u \in \mathbb{Q}_{>0}$ minutes *if and only if* the machine executes jobs of at least two different classes. For each class $c \in \{1, 2, \dots, q\}$, an extra revenue of $v_c \in \mathbb{Q}_{>0}$ euros is obtained *if and only if* all the jobs of the class are executed. The machine does not execute jobs in parallel. The company wants to choose the jobs to execute maximising the total revenue.

Problem 7.7 — total tardiness

A company needs to process $n \in \mathbb{Z}_{\geq 1}$ jobs on a single machine. For each job $j \in \{1, 2, \dots, n\}$, $t_j \in \mathbb{Q}_{>0}$ is the processing time in minutes and $d_j \in \mathbb{Q}_{>0}$ the due date in minutes. The tardiness of job j is $\tau_j = \max\{0, \kappa_j - d_j\}$, where κ_j is the completion time. The machine processes one job at a time. The company wants a sequence of the jobs that minimises the total tardiness.

Location and coverage

Problem 8.1 — capacitated facility location

A company must serve $n \in \mathbb{Z}_{\geq 1}$ clients with its product and has identified $m \in \mathbb{Z}_{\geq 1}$ candidate locations where a facility can be installed. For each client $c \in \{1, 2, \dots, n\}$, the value $d_c \in \mathbb{Q}_{>0}$ is the demand, in liters of product, of that client. For each location $l \in \{1, 2, \dots, m\}$ and each client c , the value $t_{lc} \in \mathbb{Q}_{>0}$ is the cost, in euros, of transporting one liter of product from location l to client c . For each location l , the value $u_l \in \mathbb{Q}_{>0}$ is the capacity, in liters, of the facility installed there, and $i_l \in \mathbb{Q}_{>0}$ is the installation cost. The company wants to decide where to install facilities and how to serve the clients, at minimum total cost.

Problem 8.2 — p-median

A company must choose at most $k \in \mathbb{Z}_{\geq 1}$ locations, among $m \in \mathbb{Z}_{\geq 1}$ candidates, where to open a service point, and assign each of the $n \in \mathbb{Z}_{\geq 1}$ clients to the most convenient open location. For each location $l \in \{1, 2, \dots, m\}$ and each client $c \in \{1, 2, \dots, n\}$, the value $d_{lc} \in \mathbb{Q}_{>0}$ is the distance (or service cost) between location l and client c . Every client must be assigned to exactly one open location. The company wants to decide which locations to open and how to assign clients, minimizing the sum of the distances between each client and the location it is assigned to.

Problem 8.3 — coverage with interference

An operator must choose at most $k \in \mathbb{Z}_{\geq 1}$ locations, among $m \in \mathbb{Z}_{\geq 1}$ candidates, where to install an antenna, to serve $n \in \mathbb{Z}_{\geq 1}$ clients. For each location $l \in \{1, 2, \dots, m\}$ and each client $c \in \{1, 2, \dots, n\}$, the value $s_{lc} \in \mathbb{Q}_{\geq 0}$ is the signal strength received by client c if location l is installed. A client c is *covered* if and only if the total signal strength received from all installed locations is at least the threshold $t \in \mathbb{Q}_{>0}$ and at most one installed location generates for that client a signal greater than or equal to the interference limit $b \in \mathbb{Q}_{>0}$. For each client c , the value $p_c \in \mathbb{Q}_{>0}$ is the profit if the client is covered. The operator wants to decide which locations to install to maximize the total profit.

Problem 8.4 — hub with maximum connection cost

In a computer network there are $n \in \mathbb{Z}_{\geq 1}$ terminals, each of which must be connected to exactly one hub. There are $m \in \mathbb{Z}_{\geq 1}$ hubs available, and each hub can serve at most

$k \in \mathbb{Z}_{\geq 1}$ terminals. A hub must be activated before it can serve any terminals: for each hub $j \in \{1, 2, \dots, m\}$, the value $f_j \in \mathbb{Q}_{\geq 0}$ is the activation cost. For each terminal $i \in \{1, 2, \dots, n\}$ and each hub j , the value $c_{ij} \in \mathbb{Q}_{\geq 0}$ is the cost of connecting terminal i to hub j . The objective is to decide how to assign terminals to hubs to minimize the sum of the activation costs of the hubs used and the maximum connection cost of each hub, subject to the constraints described.

Production planning

Problem 9.1 — lot sizing with a fixed cost

A company must plan the production of a single product over $n \in \mathbb{Z}_{\geq 1}$ periods. For every period $t \in \{1, 2, \dots, n\}$, the value $d_t \in \mathbb{Q}_{\geq 0}$ is the demand to be met, $p_t \in \mathbb{Q}_{> 0}$ the cost of producing one unit and $q_t \in \mathbb{Q}_{\geq 0}$ the fixed cost to be paid if production is started in that period. For every period $t \in \{1, 2, \dots, n-1\}$, the value $h_t \in \mathbb{Q}_{\geq 0}$ is the cost of keeping one unit in stock from the end of period t to the start of the next one. The initial inventory is $r_0 \in \mathbb{Q}_{\geq 0}$ and the one required at the end of the horizon is $r_n \in \mathbb{Q}_{\geq 0}$. The company wants to decide how much to produce in every period so as to meet all the demand at minimum total cost.

Problem 9.2 — production and workforce

A shoe factory must plan its production over $n \in \mathbb{Z}_{\geq 1}$ months. For every month $t \in \{1, 2, \dots, n\}$, the value $d_t \in \mathbb{Z}_{\geq 0}$ is the demand in pairs and $p_t \in \mathbb{Q}_{> 0}$ the cost of the raw materials for one pair. For every month $t \in \{1, 2, \dots, n-1\}$, the value $h_t \in \mathbb{Q}_{\geq 0}$ is the cost of keeping one pair in stock at the end of the month. At the start of the horizon the company has $m_0 \in \mathbb{Z}_{\geq 0}$ workers; every worker works $r \in \mathbb{Q}_{> 0}$ hours a month and costs $w \in \mathbb{Q}_{> 0}$ euros a month, and producing one pair requires $g \in \mathbb{Q}_{> 0}$ labour hours. At the start of every month workers may be hired, at a cost of $u \in \mathbb{Q}_{\geq 0}$ euros each; nobody is laid off. The company wants to decide how much to produce and how many workers to employ in every month, at minimum total cost.

Problem 9.3 — vehicles with minimum lot and bonus

A company produces $s \in \mathbb{Z}_{\geq 1}$ types of vehicle using $k \in \mathbb{Z}_{\geq 1}$ resources. For every resource $i \in \{1, 2, \dots, k\}$ and every type $j \in \{1, 2, \dots, s\}$, the value $a_{ij} \in \mathbb{Q}_{\geq 0}$ is the amount of resource i needed for one unit of type j , and $b_i \in \mathbb{Q}_{> 0}$ is the availability of resource i . For every type j , the value $\bar{p}_j \in \mathbb{Q}_{> 0}$ is the profit of one unit and $\bar{q}_j \in \mathbb{Z}_{\geq 1}$ the minimum quantity to be produced if that type is chosen. If the production includes at least two different types, the company collects a bonus $\bar{r} \in \mathbb{Q}_{> 0}$ (a grant for diversification). The company wants to maximise the total profit.

Mixed models

Problem 10.1 — prizes with two modes

A loyalty programme offers $s \in \mathbb{Z}_{\geq 1}$ prizes and a customer has $p \in \mathbb{Q}_{>0}$ points. Every prize $i \in \{1, 2, \dots, s\}$ can be obtained in two alternative ways: with points only, spending $a_i \in \mathbb{Q}_{>0}$ of them; or spending only $b_i \in \mathbb{Q}_{>0}$ points (with $b_i < a_i$) and adding a cash contribution of $c_i \in \mathbb{Q}_{\geq 0}$ euros. Every prize can be obtained at most once, and in one mode only. Every prize carries a preference value $d_i \in \mathbb{Q}_{>0}$, and the customer wants an overall preference of at least $\ell \in \mathbb{Q}_{>0}$. The customer wants to minimise the money spent.

Problem 10.2 — combinatorial auction

An auctioneer has a set $S = \{1, 2, \dots, n\}$ of $n \in \mathbb{Z}_{\geq 1}$ items to sell and has received $r \in \mathbb{Z}_{\geq 1}$ bids. For every bid $j \in \{1, 2, \dots, r\}$, the set $B_j \subseteq S$ is the subset of items requested and $p_j \in \mathbb{Q}_{>0}$ the profit in euros if the bid is accepted. Each item can be sold at most once, and a bid can be accepted only if all its items are available. The auctioneer wants to select a set of bids of maximum total profit.

Problem 10.3 — diet with variety

A nutritionist must compose a monthly diet choosing among $s \in \mathbb{Z}_{\geq 1}$ foods and controlling $r \in \mathbb{Z}_{\geq 1}$ nutrients. For every food $i \in \{1, 2, \dots, s\}$, the value $w_i \in \mathbb{Q}_{>0}$ is the cost of one kilo; for every food i and every nutrient $j \in \{1, 2, \dots, r\}$, the value $g_{ij} \in \mathbb{Q}_{\geq 0}$ is the amount of nutrient j contained in one kilo of food i . For every nutrient j , the monthly intake must lie between $a_j \in \mathbb{Q}_{>0}$ and $b_j \in \mathbb{Q}_{>0}$. If a food enters the diet, at least $c_i \in \mathbb{Q}_{>0}$ kilos of it must be consumed and at most $d_i \in \mathbb{Q}_{>0}$. The diet must include at least $t \in \mathbb{Z}_{\geq 1}$ different foods. The goal is the diet of minimum cost.

Problem 10.4 — trees and lights

The city administration wants to decorate $q \in \mathbb{Z}_{\geq 1}$ trees for the holidays. Every tree can be decorated according to one of the $n \in \mathbb{Z}_{\geq 1}$ possible configurations, each characterised by lights of $m \in \mathbb{Z}_{\geq 1}$ different colours. For every configuration $c \in \{1, 2, \dots, n\}$ and every colour $l \in \{1, 2, \dots, m\}$, the value $u_{cl} \in \mathbb{Z}_{\geq 0}$ is the number of lights of colour l required by configuration c , and $i_c \in \mathbb{Q}_{>0}$ is the installation cost of a tree decorated according to it. All the lights must be bought on the market, where they are sold in boxes of $k \in \mathbb{Z}_{\geq 1}$ types: for

every type $b \in \{1, 2, \dots, k\}$ and every colour l , the value $v_{bl} \in \mathbb{Z}_{\geq 0}$ is the number of lights of colour l contained in a box of type b , and $p_b \in \mathbb{Q}_{\geq 0}$ is its cost. To ensure a pleasant visual variety at least $f \in \mathbb{Z}_{\geq 1}$ different configurations must be used. The goal is to decorate the trees at minimum total cost.

Problem 10.5 — shipments in boxes

A company needs to organise the shipments from its $n \in \mathbb{Z}_{\geq 1}$ production plants to its $m \in \mathbb{Z}_{\geq 1}$ customers. The company produces $k \in \mathbb{Z}_{\geq 1}$ product types. Products are shipped in boxes: every box travels directly from one plant to one customer and has maximum capacity $w \in \mathbb{Z}_{\geq 1}$, expressed as the maximum number of product units it can carry. For every product $p \in \{1, 2, \dots, k\}$ and every customer $c \in \{1, 2, \dots, m\}$, the value $d_{pc} \in \mathbb{Z}_{\geq 0}$ is the number of units ordered. For every product p and every plant $s \in \{1, 2, \dots, n\}$, the value $a_{ps} \in \mathbb{Z}_{\geq 0}$ is the number of units available. The company wants a shipment plan that minimises the total number of boxes used.

Problem 10.6 — summer camps

A company runs $r \in \mathbb{Z}_{\geq 1}$ summer camps to host children during the holidays. For every camp $j \in \{1, 2, \dots, r\}$, the value $d_j \in \mathbb{Z}_{\geq 1}$ is the maximum number of children that can be hosted. The company has received requests from children of $s \in \mathbb{Z}_{\geq 1}$ different nationalities: for every nationality $i \in \{1, 2, \dots, s\}$ there are $f_i \in \mathbb{Z}_{\geq 0}$ girls and $g_i \in \mathbb{Z}_{\geq 0}$ boys. In every camp the number of girls must be greater than or equal to the number of boys, and the number of children of nationality $c \in \{1, 2, \dots, s\}$ must be greater than or equal to that of every other nationality. The company wants to maximise the total number of children accepted.

Problem 10.7 — antitrust split

A company has $s \in \mathbb{Z}_{\geq 1}$ branches and sells $r \in \mathbb{Z}_{\geq 1}$ products. For every branch $i \in \{1, 2, \dots, s\}$ and every product $j \in \{1, 2, \dots, r\}$, the value $v_{ij} \in \mathbb{Q}_{\geq 0}$ is the revenue in millions of euros generated by that branch with that product. Because of a new antitrust regulation the company must split into two smaller companies; every branch is indivisible and must be assigned to exactly one of the two. The company wants to partition the branches minimising the maximum difference, over all the products, between the revenues of the two new companies.

Problem 10.8 — songs on CDs

A music production company publishes a collection of $n \in \mathbb{Z}_{\geq 1}$ songs. For every song $i \in \{1, 2, \dots, n\}$, the value $d_i \in \mathbb{Q}_{\geq 1}$ is the duration in minutes. The collection must be distributed over $m \in \mathbb{Z}_{\geq 1}$ compact discs; every song must be assigned to exactly one CD, and every CD $j \in \{1, 2, \dots, m\}$ must contain at least $w_j \in \mathbb{Z}_{\geq 1}$ songs. The company wants to minimise the difference between the total duration of the longest CD and that of the shortest one.

Problem 10.9 — books on shelves

A library needs to arrange $n \in \mathbb{Z}_{\geq 1}$ books on shelves. Every book $b \in \{1, 2, \dots, n\}$ has width $w_b \in \mathbb{Z}_{\geq 1}$ and height $h_b \in \mathbb{Z}_{\geq 1}$. There are $m \in \mathbb{Z}_{\geq 1}$ shelves available, each of maximum width $c \in \mathbb{Q}_{>0}$. Every book must be assigned to exactly one shelf, and the total width of the books on a shelf cannot exceed c . The height of a shelf is that of the tallest book placed on it. The library wants to minimise the sum of the heights of the shelves.

Problem 10.10 — the depot's week

A depot supplies three shops for five days. Each day each shop has a known requirement, in pallets. The depot has an initial stock and can receive pallets from the factory with a daily order: an order costs a fixed amount to activate the transport, plus an amount per pallet. Pallets not delivered stay in the depot, and the end-of-day stock costs. Every delivery to a shop travels on a van that carries at most a given number of pallets, and every van that leaves costs. On a given day the depot cannot send out more than a given number of vans in total. The minimum-cost weekly plan is wanted.

Problem 10.11 — the technical desk

A support service covers seven days. Each day a known number of technicians must be present. Technicians are hired on weekly contracts of fixed shape: a contract states which days are worked, and has a known weekly cost. Each technician has one of three skills, and each day at least one technician of each of the three is needed. Two contracts are incompatible with the rarest skill. The minimum-cost plan is wanted.